

Sarah Canto Hyatt

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Education

PhD	University of California, Santa Barbara , Computer Science	In Progress
MS	California State University, Northridge , Computer Science	2023
BS	California State University, Northridge , Computer Science	2021
	• Minor: Mathematics	

Publications

Breccia: A Functional DSL Compiled to Egglog for Test Input Generation	2026
Peter Boyland, Sarah Canto Hyatt , Kyle Dewey, Ben Hardekopf	
ACM International Symposium on Functional and Logic Programming (FLOPS) 2026	
publication 🔗	
Mutation-based fuzzer for the Swift compiler	2026
Sarah Canto Hyatt , Kyle Dewey	
Science of Computer Programming (SCP) 2026	
publication 🔗	
Mutation-Based Fuzzing of the Swift Compiler with Incomplete Type Information	2025
Sarah Canto Hyatt , Kyle Dewey	
International Conference on Software Testing, Verification and Validation (ICST) 2025	
publication 🔗	

Research Experience

University of California, Santa Barbara , Graduate Student Researcher	Santa Barbara, CA
- Conducting research under the supervision of Ben Hardekopf, PhD. - Working at the intersection of programming languages and computer architecture, with a focus on formal verification and hardware description languages.	2024 – Present
California State University, Northridge , Research Assistant	Northridge, CA
- Conducted research under the supervision of Kyle Dewey, PhD. - Developed a type-aware mutation-based black-box fuzzer for the Swift programming language.	2022 – 2023

Teaching Experience

University of California, Santa Barbara , Teaching Assistant	Santa Barbara, CA
- Currently serving as a TA for CMPSC 8: Introduction to Computer Science (Summer'26) - Served as a TA for the following courses: - CMPSC 24: Problem Solving with Computers 2 (Spring'25, Winter'26, Spring'26) - CMPSC 160: Translation of Programming Languages (Fall'25) - CMPSC 8: Introduction to Computer Science (Winter'25 and Summer'25) - CMPSC 16: Problem Solving with Computers 1 (Fall'24)	2024 – Present

California State University, Northridge, Instructional Student Assistant

Northridge, CA
2021 – 2023

- Served as an instructional assistant for the following CS courses:
 - COMP 110/L: Introduction to Algorithms and Programming (Spring'23, Fall'23)
 - COMP 182/L: Data Structures and Program Design and Lab (Fall'22, Spring'23)
 - COMP 282: Advanced Data Structures (Fall'22)
 - COMP 333: Concepts of Programming Languages (Fall'23)
 - COMP 482: Algorithm Design and Analysis (Fall'23)
- Hosted individual and group tutoring for the following CS courses:
 - COMP 110/L: Introduction to Algorithms and Programming (Spring'21)
 - COMP 122/L: Computer Architecture and Assembly Language and Lab (Spring'21, Fall'21, Spring'22)
 - COMP 182/L: Data Structures and Program Design and Lab (Spring'21, Spring'22)
 - COMP 282: Advanced Data Structures (Spring'21)